Deepfake Audio Detector

External Audio Validation and Generalization Findings

PROGRESS REPORT 3 / OCTOBER 2026

MAIN FINDING	**The MLP performed well on held-out DFADD audio but repeatedly classified outside real-world recordings as spoofed, revealing a generalization problem that now guides the next stage of the project.**

1. Progress Since the Previous Report

The previous progress report compared Random Forest, Support Vector Machine (SVM), Gradient Boosting, Logistic Regression, and Multi-Layer Perceptron (MLP). MLP produced the strongest overall results and was selected as the final model architecture for continued development.

Since that report, the project has moved from model comparison into deployment-oriented testing. The team created a final balanced MLP training pipeline, saved the trained model for reuse, updated the prediction script to accept external audio files, and began testing audio that did not come from the DFADD dataset.

What changed in this phase

Final training set	20,000 balanced samples: 10,000 real and 10,000 spoofed.
Feature representation	20 MFCC features extracted from each recording.
Final model	StandardScaler + MLP neural network with hidden layers of 64 and 32 units.
Saved model	The trained pipeline was exported so new audio can be classified without retraining.
New evaluation	Real and AI-generated audio from outside DFADD was introduced to the model.

2. Final Balanced MLP Training

A new final-training script was created to deliberately collect equal numbers of real and spoofed samples instead of taking a mixed batch and discarding the larger class afterward. The final dataset contained exactly 10,000 real and 10,000 spoofed recordings, giving 20,000 total samples and 20 MFCC features per sample.

The data was split 80/20, resulting in 16,000 training samples and 4,000 testing samples. The model used a StandardScaler followed by an MLPClassifier. The preprocessing scaler and neural network were saved together as one reusable pipeline.

Metric	Result
Accuracy	95.65%
Test samples	4,000
Correct predictions	3,826
False positives	96 real samples labeled spoofed
False negatives	78 spoofed samples labeled real

Internal DFADD Evaluation

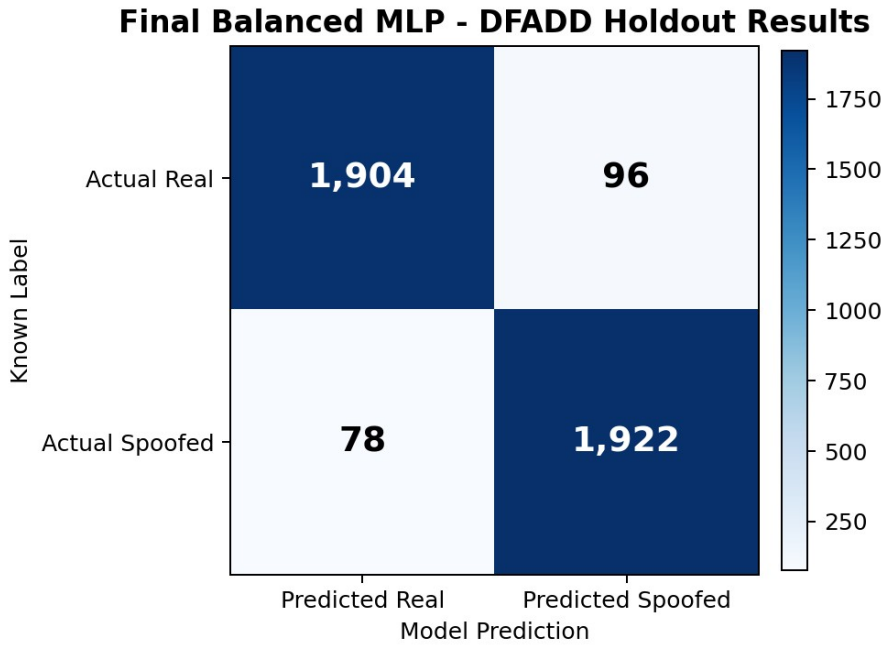


Figure 1. Confusion matrix for the final balanced MLP on the 4,000-sample DFADD holdout set.

3. External Audio Testing

The next goal was to determine whether the detector could classify audio that was completely outside the training database. The prediction pipeline converts incoming audio to mono at 16 kHz, extracts the same 20 MFCC features used during training, and passes those features to the saved MLP model. This allows recordings from devices such as an iPhone or Mac microphone to be tested directly.

Preliminary testing showed a clear difference between performance on DFADD and performance on outside recordings. The model repeatedly predicted outside audio as spoofed, including real human recordings. A known fake recording was also classified as spoofed, meaning the detector could still recognize some fake audio but was not reliably separating real and fake audio in the new recording environment.

Preliminary external examples

Sample	Known Label	Model Prediction	Result
Test 2	Real	Spoofed	**Incorrect**

Sample	Known Label	Model Prediction	Result
Test 3	Fake / AI	Spoofed	**Correct**
Test 4	Real	Spoofed	**Incorrect**

Additional recordings made with Mac and iPhone microphones showed the same general behavior: real speech was frequently classified as spoofed. Some incorrect predictions were returned with very high confidence, showing that model confidence should not be interpreted as proof that the classification is correct.

NOTE The outside-audio test is preliminary and uses a small number of recordings, so it should not be treated as a full external accuracy measurement. Its value is that it exposed a generalization weakness that was not visible when testing only on DFADD.

4. Generalization Finding

The current evidence suggests that the MLP learned useful patterns for distinguishing real and spoofed speech within DFADD, but some of those patterns do not transfer well to audio recorded under different conditions. This is a generalization, or domain-shift, problem.

Possible sources of domain shift include microphone characteristics, recording environment, compression, background noise, speaker differences, and the specific synthesis methods used to generate the spoofed audio in DFADD. Because both Mac and iPhone recordings showed similar behavior, the issue does not appear to be limited to a single microphone.

This result changes the interpretation of the project in an important way. High performance on a held-out portion of one dataset does not automatically mean the detector will perform equally well on real-world audio from a different source. External testing is therefore necessary before claiming that the system can generalize beyond its training data.

5. Methodological Improvement

Reviewing the earlier 20,000-sample experiment also revealed that the old balancing method did not actually train on 20,000 balanced samples. That run initially streamed 20,000 recordings, but because the dataset contained many more spoofed than real samples, balancing reduced the usable set to 8,062 samples: 4,031 real and 4,031 spoofed.

The new final-training pipeline corrects this by continuing to collect samples until it reaches 10,000 real and 10,000 spoofed recordings. This produces a true 20,000-sample balanced training dataset and makes the final model-development process more controlled and reproducible. The resulting 95.65% DFADD holdout accuracy is therefore from a different, more deliberately constructed training setup than the earlier comparison run.

6. Current Status and Next Steps

Completed	Five model families compared and MLP selected for continued development.

Completed	Final balanced 20,000-sample MLP trained, evaluated, and saved.
Completed	Prediction pipeline updated to accept audio files outside DFADD.
Current Finding	External recordings reveal poor generalization, especially real audio being mislabeled as spoofed.
Next Steps	Cross-train and evaluate with additional independent real/fake audio datasets, then repeat external testing across devices and recording conditions.

7. Progress Report Conclusion

This phase of the project moved the detector beyond internal dataset testing and into real-world validation. The final balanced MLP achieved 95.65% accuracy on unseen DFADD samples, but outside recordings revealed that the model does not yet generalize reliably to new audio sources. Real human speech was repeatedly classified as spoofed, even when recorded on different devices.

Rather than treating this as the end of the project, the result identifies the most important remaining research problem. The next phase will focus on broader datasets and cross-dataset evaluation so the detector learns characteristics of real and AI-generated speech that transfer across recording environments. This external-validation result provides a clearer and more realistic measure of what the current system can and cannot do.

NEXT RESEARCH PRIORITY	**Strong performance inside one dataset is not enough. The next milestone is to improve cross-dataset generalization and show that the detector can distinguish real and AI-generated speech across different recording environments.**